MIL-T-12270C
13 March 1978
SUPERSEDING
MIL-T-12270B
1 June 1970

MILITARY SPECIFICATION

TIE-PLATES, RAILWAY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers steel tie-plates used in railroad track in maintaining track alinement and as a protection for the ties against mechanical wear.

1.2 Classification. Tie-plates shall be for the following rail weights, as specified:

60 pound (27 kg) ASCE rail.
70 pound (32 kg) ASCE rail.
90 pound (41 kg) ARA-A rail.
115 pound (52 kg) AREA rail.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 2250

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document

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SPECIFICATIONS

FEDERAL

QQ-W-461 - Wire, Steel, Carbon, (Round, Bare and Coated).

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN RAILWAY ENGINEERING ASSOCIATION (AREA)

Specification for Low Carbon or Hot-Worked, High Carbon Steel Tie-Plates.
AREA Manual, Vol. I, Construction and Maintenance Section.

(Application for copies should be addressed to the American Railway Engineering Association, 59 East Van Buren Street, Chicago, IL 60605.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street NW, Washington, DC 20036.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 67 - Low Carbon Steel Tie-Plates.
A 241 - Hot-Worked High-Carbon Steel Tie-Plates.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Description. The tie-plates shall be fabricated in accordance with American Railway Engineering Association (AREA) specification, and ASTM A 67 and A 241 as applicable for low-carbon or hot-worked, high carbon steel tie-plates.

3.2 Material. Steel shall be as specified in AREA specifications (see 6.3).

3.3 Identification marking. Tie-plates shall be identified in accordance with AREA specifications.

3.4 Workmanship. Workmanship for fabrication and finish of tie-plates shall be in accordance with AREA specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

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4.3 Quality conformance inspection.

4.3.1 Examination. The tie-plate shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The tie-plate shall be tested as specified in 4.4.2. Failure of any test shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. Tie-plates shall be examined for the following defects:

101. Tie-plates for specified rail weight not as requested (see 1.2).
102. Tie-plates not fabricated as specified (see 3.1).
103. Material not as specified (see 3.2).
104. Identification markings not as specified (see 3.3).
105. Workmanship not as specified (see 3.4).

4.4.2 Tests. Chemical composition and tests shall be in accordance with AREA specification for steel tie-plates.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Examination. The tie-plates shall be examined for the following defects:

106. Packing not Level A or Commercial as specified (see 5.1).
107. Packing for Level A not as specified (see 5.1.1).
108. Packing for Commercial not as specified (see 5.1.2).
109. Marking for military level of packaging not in accordance with MIL-STD-129 (see 5.2.1).
110. Marking for Commercial level of packaging not as specified (see 5.2.2).

5. PACKAGING

5.1 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

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5.1.1 Level A. Tie-plates of like description shall be secured together, in quantities not to exceed 150 pounds (68 kg) gross weight, with 50 feet annealed wire conforming to QQ-W-461, Finish 5, Class 2, composition, tensile-strength and wire size as applicable.

5.1.2 Commercial. Tie-plates of like description shall be prepared for shipment in a manner to assure carrier acceptance and delivery to destination, without damage, at the lowest ratings of the carrier in compliance with Uniform Freight Classification rules or National Motor Freight classification rules.

5.2 Marking.

5.2.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.2.2 Commercial packaging. Marking shall reflect as a minimum, the item name, National Stock Number, quantity, contract number, and appropriate address markings. The marking shall be applied with a waterproof material, be legible, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. Tie-plates are interposed between a rail or other track structure and a tie.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Rail weight required (see 1.2).
(c) Whether tests are required (see applicable AREA specifications (see 4.4.2.)).
(d) Degree of packing required (see 5.1).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

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MIL-T-12270C Tie-Plates, Railway

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